CarbonLedger

Scope 1, 2 and 3 Emissions Report

Meridian Electronics Manufacturing, Inc.
Reporting period: 2026-01-01 to 2026-06-30
GWP version: AR6
Track: Sustainability & Social Impact

Factor year distribution (factors used): 2025: 1, 2026: 19

Total emissions: 454.301 t CO2e

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1. Cover

Reporting entity: Meridian Electronics Manufacturing, Inc.
Reporting period: 2026-01-01 to 2026-06-30
Generated on: 2026-10-15
GWP version: AR6 (100-year global warming potentials; see chapter 3)
Track: Sustainability & Social Impact
Factor year distribution: 2025: 1 factors used, 2026: 19 factors used
Disclaimer: See chapter 10

2. Executive summary

Total emissions for the reporting period amount to 454.301 t CO2e.

Scope	t CO2e	Share
Scope 1	160.186	35.3%
Scope 2	208.784	46.0%
Scope 3	85.331	18.8%

Top emission source	t CO2e
District heat supply for office campus (2,880 GJ converted)	208.784
Refrigerant HFC-32 top-up for plant HVAC	148.8
Employee commuting rail season passes, aggregate H1 2026	34.841

1. Prioritize the largest emission sources first: District heat supply for off
HFC-32 top-up for plant HVAC; Employee commuting rail season passe
the majority of the reported footprint, so procurement and logistics actio
reductions per unit of effort.
2. Adopt supplier-specific data collection for the top suppliers identified
factors progressively; each replaced factor lowers the uncertainty level
3. Institutionalize quarterly re-runs of this accounting pipeline so that ne
ledger, keeping the report current and allowing reduction progress to be

3. Accounting method and boundary

This report follows the GHG Protocol Corporate Value Chain (Scope 3) Accounting and Reporting Standard in conjunction with the Corporate Standard. Organizational boundary: operational control. Operational boundary: Scope 1 direct, Scope 2 purchased energy, and Scope 3 value-chain categories as identified per record.

GWP version: AR6, sourced from IPCC assessment report 100-year values. Cross-GWP-version factor usage: 20 of 20 records (100.0%) (each flagged in the interface and downgraded to at least medium uncertainty).

Factor data year	Factors used
2025	1
2026	19

Data quality overall assessment: Data quality is moderate: 20 of 20 reco
version different from the task setting and are flagged accordingly.

4. Emissions breakdown

Scope	t CO2e
Scope 1	160.186
Scope 2	208.784
Scope 3	85.331

Scope 3 category	t CO2e
Cat-1 Purchased goods and services	30.468
Cat-2 Capital goods	5.862
Cat-3 Fuel- and energy-related activities	9.252
Cat-4 Upstream transportation and distribution	2.48
Cat-5 Waste generated in operations	0.02
Cat-6 Business travel	2.409
Cat-7 Employee commuting	34.841
Cat-8 Upstream leased assets	0
Cat-9 Downstream transportation and distribution	0
Cat-10 Processing of sold products	0
Cat-11 Use of sold products	0
Cat-12 End-of-life treatment of sold products	0
Cat-13 Downstream leased assets	0
Cat-14 Franchises	0
Cat-15 Investments	0

Supplier (top 10)	t CO2e
Valero	11.386
Lone Star HVAC Services	148.8
Austin Energy	0
Austin District Energy	208.784
Aluworks Supply	27.924
Unspecified	47.265
Austin PCB Fab	0.003
Heartland Machine Tools	5.862
Maersk	2.191
Central Texas Environmental Services	0.019

5. Activity data detail

Source	Description	Quantity	Factor	t CO2e	Uncertainty
doc_ac950e36#page=1	Diesel fuel (average biofuel blend) for own fleet, combustion	14000 kg	EF-D26-WTTFUELS-015	11.386	high
doc_d7770e08#page=1	Refrigerant HFC-32 top-up for plant HVAC	12 kg	EF-D26-REFRIGERANTOTHER-004	148.8	medium
doc_067eeb1a#page=1	Factory grid electricity, H1 2026	1250000 kWh	EF-D26-BUSINESSTRAVELAIR-018	0	high
doc_6fce52b4#page=1	District heat supply for office campus (2,880 GJ converted)	799997.76 kWh	EF-E25-FUEL-045	208.784	high
doc_2b7da642#page=1	Aluminium sheet 5052 for chassis panels	9500 kg	EF-D26-FUELS-007	27.924	high
doc_e03d6fe3#page=1	PET film, food grade	1200 kg	EF-D26-WASTEDISPOSAL-113	0.006	high
doc_e03d6fe3#page=1	PP injection pellets	2600 kg	EF-D26-WASTEDISPOSAL-010	0.003	high
doc_e03d6fe3#page=1	LDPE packaging film	900 kg	EF-D26-WASTEDISPOSAL-111		

				0.008	high
doc_1d92d428#page=1	4-layer PCB assemblies (12 lots)	640 kg	EF-D26-WASTEDISPOSAL-006	0.003	high
doc_592506fd#page=1	CNC machining centre VM-850 (capital equipment)	8200 kg	EF-D26-WTTFUELS-031	5.862	high
doc_6fd9add0#page=1	Well-to-tank upstream emissions of fleet diesel (matches D-01 load)	14000 kg	EF-D26-WTTFUELS-017	9.252	high
doc_37eada2e#page=1	FCL ocean freight of sensor modules, Houston to Hamburg	17300 kg	EF-D26-SEAFREIGHT-034	2.191	high
doc_c9cc4c04#page=1	Inbound road freight, Dallas to Austin (ambient rigid truck)	6800 kg	EF-D26-WTTROADFREIGHT-007	0.198	high
doc_761c2921#page=1	Landfill disposal of mixed production waste	4200 kg	EF-D26-WASTEDISPOSAL-080	0.02	high
doc_0a15e5d8#page=1	Return economy flight AUS-HAJ for trade fair, 1 passenger	16900 km	EF-D26-BUSINESSTRAVELAIR-019	2.409	high
doc_498f0e48#page=1	Heat-treated wooden export pallets (ISPM-15)				

		520 kg	EF-D26-MATERIALUSE-070	0.545	high	
doc_40ba16d4#page=1						
	Employee commuting rail season passes, aggregate H1 2026	210000 km	EF-D26-BUSINESSTRAVELLAND-013	34.841	medium	
doc_88b783c9#page=1						
	International express courier, Austin to London	85 kg	EF-D26-WTTAIRFREIGHT-004	0.091	high	
doc_de7b95d6#page=1						
	Toughened display glass panels	2400 kg	EF-D26-MATERIALUSE-026	1.976	high	
doc_86cf34f7#page=1						
	A4 copy paper and toner cartridges	780 kg	EF-D26-WASTEDISPOSAL-096	0.004	high	

6. Uncertainty statement

Level	Records
low	0
medium	2
high	18

Main uncertainty sources: document extraction confidence on scanned
factors rather than supplier-specific data, and unit conversions between

1. Request primary activity data and factor references from the three lar
2. Standardize document formats with recurring logistics providers so ex
3. Complete the missing required fields currently parked in the review q
resolved.

7. Reduction scenario advice

Scenario	Reduction t CO2e	Reduction %	
Transport mode shift	0.112	0.02%	Transport mode shift: replacing air and express legs cutting 0.112 t CO2e (0.02% of the baseline). Assu UK Government conversion factor set and cover eq replacement ratio can be tuned between 0% and 10 actual feasibility depends on route schedules, servi this tool cannot verify; cold-chain and time-critical fl averages, so site-specific results will differ; seconda of scope.

Scenario	Reduction t CO2e	Reduction %	
Supplier swap	100.453	22.11%	Supplier swap: re-sourcing the activity of the three h HVAC Services, Unspecified) to the lowest-value fa 100.453 t CO2e. Assumptions: the replacement fact factor set and cover equivalent service levels; activi between 0% and 100% and only the replaced share schedules, service contracts, supplier availability an and time-critical flows may not tolerate slower mode differ; secondary effects such as packaging change

Scenario	Reduction t CO2e	Reduction %	
Material substitution	0.805	0.18%	Material substitution: switching high-emission mater the same material family for 1 record(s), abating 0.8 are kept unchanged and listed individually. Assumpt UK Government conversion factor set and cover eq replacement ratio can be tuned between 0% and 10 actual feasibility depends on route schedules, servi this tool cannot verify; cold-chain and time-critical fl averages, so site-specific results will differ; seconda of scope.

8. Appendix I — Emission factors used

ID	Name	Value	Unit	Year	Source
EF-D26-WTTFUELS-015	Fuel upstream (WTT) — Liquid fuels / Aviation spirit — Well-to-tank (t)	813.26357	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-REFRIGERANTOTHER-004	Refrigerant — Kyoto protocol products / HFC-23 — Refrigerant leak (kg)	12400	kgCO2e/kg	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-BUSINESSTRAVELAIR-018	Business travel air — Flights / First class — Flights excl. radiative forcing (RF) (p.km)	0.27701	kgCO2e/p.km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-E25-FUEL-045	Fuel — Other Oil (>401 deg F) (Stationary Combustion, per MMBtu)	76.46300000000001	kgCO2e/MMBtu	2025	US EPA GHG Emission Factors Hub, January 2025
EF-D26-FUELS-007	Fuel — Gaseous fuels / LPG — Combustion (t)	2939.36095	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-113	Waste — Plastic /				

	Plastics: PET (incl. forming) — Disposal: closed-loop recycling (t)	4.65358	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-010	Waste — Construction / Asphalt — Disposal: landfill (t)	1.27043	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-111	Waste — Plastic / Plastics: LDPE and LLDPE (incl. forming) — Disposal: landfill (t)	9.00687	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-006	Waste — Construction / Average construction — Disposal: combustion (t)	4.65358	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WTTFUELS-031	Fuel upstream (WTT) — Liquid fuels / Marine fuel oil — Well-to-tank (t)	714.86545	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WTTFUELS-017	Fuel upstream (WTT) — Liquid fuels / Burning oil — Well-to-tank (t)	660.82605	kgCO2e/t		

				2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-SEAFREIGHT-034	Freight — Cargo ship / 0–4999 dwt — Sea shipping (direct combustion) (t.km)	0.01407	kgCO2e/t.km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WTTROADFREIGHT-007	Freight upstream (WTT) — WTT- HGV (non-refrigerated, all diesel) / Rigid (>7.5 tonnes-17 tonnes) — Diesel (t.km)	0.09246	kgCO2e/t.km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-080	Waste — Metal / Metal: mixed cans — Disposal: open-loop recycling (t)	4.65358	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-BUSINESSTRAVELAIR-019	Business travel air — Flights / International, to/from non-UK / Average passenger — Flights incl. radiative forcing (RF) (p.km)	0.14253	kgCO2e/p.km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-MATERIALUSE-070	Materials — Paper / Paper and board: paper — Recycled material, closed-loop (t)	1048.59114	kgCO2e/t	2026	UK Government GHG Conversion

					Factors for Company Reporting 2026 (DESNZ)
EF-D26-BUSINESSTRAVELLAND-013	Business travel land — Cars (by size) / Average car — Fleet average fuel (unknown mix) (km)	0.16591	kgCO2e/km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WTTAIRFREIGHT-004	Freight upstream (WTT) — WTT-freight flights / International, to/from non-UK — WTT freight flights (upstream fuel production) (t.km)	0.13516	kgCO2e/t.km	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-MATERIALUSE-026	Materials — Other / Glass — Recycled material, closed-loop (t)	823.18954	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)
EF-D26-WASTEDISPOSAL-096	Waste — Plastic / Plastics: average plastic film — Disposal: open-loop recycling (t)	4.65358	kgCO2e/t	2026	UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ)

9. Appendix II — Citation index

Data point	t CO2e	Reference
Total emissions	454.301	calculation-set:task_792e659e
Scope breakdown: Scope 1	160.186	aggregation:task_792e659e (Scope 1)
Scope breakdown: Scope 2	208.784	aggregation:task_792e659e (Scope 2)
Scope breakdown: Scope 3	85.331	aggregation:task_792e659e (Scope 3)
Record rec_72ab5d48: Diesel fuel (average biofuel blend) for own fleet, combustion	11.386	record:rec_72ab5d48 (doc_ac950e36#page=1)
Record rec_a2501d04: Refrigerant HFC-32 top-up for plant HVAC	148.8	record:rec_a2501d04 (doc_d7770e08#page=1)
Record rec_87d08d9e: Factory grid electricity, H1 2026	0	record:rec_87d08d9e (doc_067eeb1a#page=1)
Record rec_59c6020b: District heat supply for office campus (2,880 GJ converted)	208.784	record:rec_59c6020b (doc_6fce52b4#page=1)
Record rec_5dff8874: Aluminium sheet 5052 for chassis panels	27.924	record:rec_5dff8874 (doc_2b7da642#page=1)
Record rec_0d62e0d9: PET film, food grade	0.006	record:rec_0d62e0d9 (doc_e03d6fe3#page=1)
Record rec_8a3792ec: PP injection pellets	0.003	record:rec_8a3792ec (doc_e03d6fe3#page=1)
Record rec_c922e177: LDPE packaging film	0.008	record:rec_c922e177 (doc_e03d6fe3#page=1)
Record rec_fa7e3731: 4-layer PCB assemblies (12 lots)	0.003	record:rec_fa7e3731 (doc_1d92d428#page=1)
Record rec_d4e9ac1d: CNC machining centre VM-850 (capital equipment)	5.862	record:rec_d4e9ac1d (doc_592506fd#page=1)
Record rec_50e80310: Well-to-tank upstream emissions of fleet diesel (matches D-01 load)	9.252	record:rec_50e80310 (doc_6fd9add0#page=1)
Record rec_1d9f9d15: FCL ocean freight of sensor modules, Houston to Hamburg	2.191	record:rec_1d9f9d15 (doc_37eada2e#page=1)
Record rec_766ec89b: Inbound road freight, Dallas to Austin (ambient rigid truck)		

	0.198	record:rec_766ec89b (doc_c9cc4c04#page=1)
Record rec_e1eb3031: Landfill disposal of mixed production waste	0.02	record:rec_e1eb3031 (doc_761c2921#page=1)
Record rec_e3d7b7ff: Return economy flight AUS-HAJ for trade fair, 1 passenger	2.409	record:rec_e3d7b7ff (doc_0a15e5d8#page=1)
Record rec_3f51f90b: Heat-treated wooden export pallets (ISPM-15)	0.545	record:rec_3f51f90b (doc_498f0e48#page=1)
Record rec_f6030aec: Employee commuting rail season passes, aggregate H1 2026	34.841	record:rec_f6030aec (doc_40ba16d4#page=1)
Record rec_361a46ed: International express courier, Austin to London	0.091	record:rec_361a46ed (doc_88b783c9#page=1)
Record rec_c04d7b6e: Toughened display glass panels	1.976	record:rec_c04d7b6e (doc_de7b95d6#page=1)
Record rec_adb8d497: A4 copy paper and toner cartridges	0.004	record:rec_adb8d497 (doc_86cf34f7#page=1)
S-1 reduction	0.112	scenario:task_792e659e_S-1
S-2 reduction	100.453	scenario:task_792e659e_S-2
S-3 reduction	0.805	scenario:task_792e659e_S-3

10. Disclaimer

This tool output does not constitute an audit opinion or legal advice. Users are advised to perform their own verification before external disclosure.

Excluded record	Description	Reason